FARAYA BAIG

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EDUCATION

Bachelor's in Computer Science
FAST National University of Computer & Emerging Sciences
Aug 2019 – Jun 2023

SKILLS

Programming: Python, C++, Object Oriented Programming, Design Patterns

Agentic Frameworks: AutoGen, CrewAI, LangChain, Google ADK, OpenAI Agents SDK, MCP

AI / ML: NLP & NLU, LLMs, STT/TTS, Multi-Agent Systems, RAG, Prompt Engineering, Finetuning LLMs, AI Agents & Tool Calling, MLOps

AI Security: AI Infrastructure Security, Adversarial AI / Red Teaming

Frameworks: TensorFlow, PyTorch, vLLM

Data & Databases, Vector Databases (Pinecone, Qdrant, ChromaDB), PostgreSQL, MySQL

Engineering: Transformers Architecture, NumPy, Pandas, FastAPI, Scikit-learn, REST APIs

Cloud & DevOps: AWS / Cloud Infrastructure, Docker, Git, NPM

PROFILE

AI/ML Specialist with 4+ years of experience, focused on Agentic AI, intelligent automation, NLP/NLU, Large Language Models (LLM), Speech AI, and AI Security. Experienced in building AI-powered solutions that automate manual business processes, orchestrate workflows, and connect LLMs with enterprise systems and tools. Skilled in developing autonomous agents, optimizing AI systems, and securing LLM and agentic AI applications to deliver efficient, reliable business solutions.

EXPERIENCE

Ebryx

AI Software Engineer — Oct 2025 – Present

- Built an **autonomous QA agent** that self qualifies NOVA microhypervisor builds across 29 hardware configurations and cloud bare metal servers, cutting multi hour engineer supervised cycle to few minutes.
- **Shipped QA agent** as Slack bot with natural language command parsing, giving engineers self- serve test runs on demand; added budget guardrails and failure aware retries that cut cloud spend and reduced flaky reruns.
- Ran **adversarial testing** and **threat modeling** across **LLMs, agentic frameworks,** and **MCP-based systems** for BlueRock.io, uncovering real attack vectors in AI pipelines before they could be exploited.
- Built **automated AI security scanning** into LLM deployments, identifying security gaps in agentic systems, replacing periodic manual review with checks on every build.

Emumba

AI Software Engineer — Mar 2025 – Sep 2025

- Architected and developed a **GenAI onboarding platform** (**Genie Mentor**) using **multi-agent orchestration** and **RAG** across internal engineering knowledge. This replaced ad hoc, tribal-knowledge onboarding with a **source-grounded, self-serve system**. Resulting in shorter onboarding cycle and reduced senior engineer support hours by over 50%.
- Designed a **multi-step agentic workflow** with planner, retriever, executor, evaluator, and editor roles over **MCP tool interfaces** to GitHub, Notion, and Confluence, unifying 3 siloed sources into one query path, resulting in 1 system to resolve engineering questions and saving considerable time.
- Fine-tuned **Qwen 2.5 3B** and **Mistral 7B** for **function calling** and **RAG tool-use** workloads, evaluating model behavior across domain-specific tasks. This identified where off-the-shelf models fell short on structured tool calls. Emumba gained smaller, cheaper models tuned to actually perform the function-calling tasks the platform needed.
- Built a systematic **LLM/RAG evaluation** and **prompt optimization** workflow using **DSPy**, using evaluation-driven iteration to refine prompts and improve the fine-tuned models' **retrieval quality** and **response grounding**. This gave a measurable, repeatable way to improve model performance instead of manual trial and error. The team gained a faster, evidence-based path to better fine-tuning results.

Afiniti

AI Software Engineer — Jul 2023 – Mar 2025

- Built **speech-to-text pipelines** around Canary-1B and Whisper, hitting under **5%-word error rate** with domain-specific validation for NER and numerical accuracy. This turned raw call audio into accurate, **structured transcripts** other teams could build on. Resulting in a reliable transcription layer feeding all downstream analytics.
- Deployed **NLP models** for intent, sentiment, emotion, summarization, and outcome/offer detection at **95%+** accuracy. This automated what had been manual call review and

reporting. The business gained scalable **customer-support analytics** without added headcount.

- Automated large-scale **data labeling** using **Mistral** and **LLaMA-3**, cutting labeling turnaround by **~50%**. This removed a major bottleneck in model development. Resulting in a faster ML development cycle.
- Built an intent-aware **conversational AI system** using **LangChain** and **Phi-3** for customer-support use cases, demoed to **Salesforce**. This showcased automated response generation for common queries to a major potential client. Afiniti gained a strong proof-point for enterprise deals in active sales conversations.
- Built a **telecom churn prediction model** at **96% accuracy** using customer behavioral data. This flagged at-risk customers before they left. The business gained a data-driven tool to support retention strategy.

CureMD Fellowship

Research & Development Aug 2022 – Apr 2023

- Built a **clinical decision support system validating cancer treatment plans** against NCCN guidelines, flagging deviations from clinical standards through an interactive machine-readable knowledge graph that doubles as a reusable knowledge base for downstream clinical AI work.

Other Projects

Natural Language to DB Query Assistant

- Built an **AI-powered assistant** using CrewAI to translate natural language queries into SQL, and execute them on a PostgreSQL database, returning structured JSON results.

Customer Feedback Portal with Sentiment Analysis

- Built a WebApp featuring an **AI chatbot** powered by Twitter-roBERTa-base for sentiment analysis on customer interactions.